

was adjusted to the point where we couldn't tell the difference between the two speakers.

Having now satisfied themselves that nobody in the audience would be disappointed, the time had come for the gentlemen to start setting the stage for the concert.

Stick To The Plan

"What's the big deal?" we asked when we witnessed some animated discussions and a few profuse apologies involving last-minute changes to the stage plan.

As Sunil put it, "Audio is a funny thing. See that curtain there?" (We did.) "If they were to open it or close it a little more, the sound we hear is going to change, and we'll have to adjust the setup even for small things like that. Instruments also sound different the moment we change their positions, so we need to fix a stage plan in advance."

It isn't just the audience—the artists need to hear themselves play as well, and "monitor" speakers are set up where they're supposed to stand. These can give a monitor engineer (the guy controlling the monitors) some memorable nightmares—if the band's microphones were to get too close to the monitor, sound from the speakers would enter the microphones, come out of the speakers, re-enter the microphone, come out of the speakers again...this is a deadly loop, and creates the horrible whine called *feedback*.

And if you think that's not complicated enough, rock guitarists use feedback to enhance their playing style and get some weird sounds—so not only do the guys have to make sure that microphones don't, they also have to make sure that the guitars do provide feedback when required.

"More artists today are starting to prefer in-ear monitors, so there's no chance of feedback there", says Sunil with an unmistakable sigh of

"Lower frequencies tend to move speaker cones a lot, so bass must go to the bigger, heavier speakers built to handle this movement"



The Crown Itech amplifier powers all the speakers



Jethro Tull

Jethro Tull was formed in Blackpool, England in 1967. Their first major success came with 1971's *Aqualung*, an album that still enthralled rock fans all over the world. Over the years, they have experimented with strange and wonderful sounds while the music community struggled to decide whether to classify them as Hard Rock, Folk Rock, Progressive Rock, Blues Rock or any other Rock, for that matter.

They are:

Ian Anderson—Flute, Guitar, Bouzouki, Mandolin, Harmonica and Vocals

Martin Barre—Electric/Acoustic Guitar, Flute

Andrew Giddings—Keyboards

Jonathan Noyce—Bass Guitar

Doane Perry—Drums and Percussion

On the web: www.jethrotull.com

relief, "and almost no Indian classical musician uses a monitor at all—they just work off the sound that the audience hears."

With separate speakers for the audience and the artists, it stands to reason that there would be two different people to cater to them—the Front Of House (FOH) engineer sits at, well, the front of the "house" (that's hall, auditorium or stadium to the rest of us), and hears what the audience hears. It's his job to ensure that everything is just so for the audience.

For their own needs, the artists turn to the monitor engineer, conveniently located near the stage, who will tweak the sound in their monitor speakers—even right in the middle of a song!

One By One

As the band begins to trickle in, blood pressures run high as everyone runs around making sure everything is ready for the Moment of Truth—the Sound Check.

We were a tad taken aback when Sunil whipped out a Tablet PC, a microphone, and a Wi-Fi router at this point. "All the equipment—the amplifiers, the signal processors—



The Goods

The Speakers: JBL VT4889 Line Array Loudspeakers—the main loudspeaker system
JBL VT4880 Line Array Subwoofers—also on the main loudspeaker system
JBL VRX 900 Line Array Loudspeakers—for the front row audience
Mackie SRM 450 Monitor Speakers—Monitors for the artists

The Mixers

Midas Verona 48 Input Console for the House
Allen & Heath 40 Input VCA Console for the Monitors

The Amplifier: The Crown Itech, which can also be fully controlled through a standard Ethernet connection.

The DSP: The XTA DP 428 and the XTA DP 226 Audio Management System, which manage the entire loudspeaker system.

The Equalisers: The Klark Teknik DN 370 Graphic Equalisers and the XTA GEQ 600 Graphic Equalisers are always on hand to cut out any feedback that might occur during the performance.

The Laptops: Acer Tablet PCs with 1.6 GHz Pentium Ms, 1 GB of RAM, and a Digigram

VXPocket Sound Card ("because it's the best—it has almost no self noise, so what it records is a pretty accurate reproduction of what is actually there")

The Software: XTA Audio Core to control the XTA Processors SIA's SmaartLive for all the tests and measurements
Crown's IQWic to control the Crown Amplifiers Gossip Corporate client for the FOH and Monitor Engineers to chat while the concert goes on

Approximate cost of aforementioned equipment (plus non-mentioned equipment): Rs. 2,40,00,000 (Rupees Two Crore, Forty Lakhs and Paise Zero Only)